

Vertex of the intersection

Lemma 3: If x is a vertex of $P(M_1) \cap P(M_2)$, then there

are chain: $\mathcal{C}: \emptyset \in C_1 \subseteq \dots \subseteq C_k$

$\mathcal{D}: \emptyset \in D_1 \subseteq \dots \subseteq D_\ell$ s.t.

(I) $\forall i, x(C_i) = r_1(C_i), \quad \forall j, x(D_j) = r_2(D_j)$

④ (II) $\{x^S : S \in \mathcal{C} \cup \mathcal{D}\}$ - set of vectors in $\mathbb{R}^E$

contains $|\text{supp}(x)|$ lin. indep. vectors.

pf: Given x , we can construct chains $\mathcal{C}$ and $\mathcal{D}$
s that property (I) holds.

Lemma 3 proof — cont —

$$P(M_1) \cap P(M_2) = \left\{ \underline{x} \geq 0, \begin{array}{l} x(u) \leq r_1(u) \\ x(u) \leq r_2(u) \end{array} \forall u \in E \right\}$$

x is a vertex: $\Rightarrow |E|$ tight lin. ind constraints.

$|E| - |\text{supp}(x)|$ come from " $x=0$ ".

And thus $|\text{supp}(x)|$ must come from: lin. indep. vectors x^T 
s.t. T is tight w.r.t. r_1 or r_2 .

$$x(u) = \underline{\langle x^u, x \rangle}$$

Lemma 2 tell us: $S_1 \triangleq \text{span} \{x^C : C \in \mathcal{C}\}$ contains any x^T
s.t. T is tight for r_1 . And: $S_2 \triangleq \text{span} \{x^D : D \in \mathcal{D}\}$, similarly.

$\Rightarrow \text{span} \{x^u : u \in \mathcal{C} \cup \mathcal{D}\} \supseteq \underline{S_1 \cup S_2}$ which contains all x^T s.t. T
is tight for r_1 or r_2

$$\Rightarrow \dim \text{span} \{x^u : u \in \mathcal{C} \cup \mathcal{D}\} \geq |\text{supp}(x)|. \quad \square$$

Proving Main Theorem

We prove: $P(M_1 \cap M_2) = P(M_1) \cap P(M_2)$ by induction.

Exercise: $P(M_1 \cap M_2) \subseteq P(M_1) \cap P(M_2)$ (follows from the def'n)

Let x a vector in $P(M_1) \cap P(M_2)$ but not in $P(M_1 \cap M_2)$.

(i) Deletion

If $x_i = 0$ for some i , delete to form $M_{1 \setminus i}$, $M_{2 \setminus i}$

And $\tilde{x} = x_{\setminus i} = x_{-i} = (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_{|K|})$.

Then: $\tilde{x} \in P(M_{1 \setminus i}) \cap P(M_{2 \setminus i}) \xrightarrow{\text{by induction}} P(M_{1 \setminus i} \cap M_{2 \setminus i})$

$\Rightarrow x = (\tilde{x}, 0) \in P(M_1 \cap M_2)$.

Proof — continued —

(ii) Contraction (Recall: $I \in \mathcal{I}/i$ iff $I+i \in \mathcal{I}$)

If $\underline{x_i = 1}$ for some i : Consider $M_{1/i}, M_{2/i}$

and $\tilde{x} = x_{-i}$.

$$\begin{aligned} \text{Thus: } \forall u \in E-i : \tilde{x}(u) &= x(u+i) - 1 \\ &\leq r_i(u+i) - 1 \\ &= r_{M_{1/i}}(u) \end{aligned}$$

$$\Rightarrow \tilde{x} \in P(M_{1/i}) \cap P(M_{2/i}) = P(M_{1/i} \cap M_{2/i})$$

$$\Rightarrow \tilde{x} = \sum \lambda_\alpha x^{I_\alpha}, \quad \lambda_\alpha \geq 0, \quad \sum \lambda_\alpha = 1.$$

$$I_\alpha \in \mathcal{I}_{1/i} \cap \mathcal{I}_{2/i} \Leftrightarrow (I_\alpha + i) \in \tilde{\mathcal{I}}_1 \cap \tilde{\mathcal{I}}_2.$$

$$\Rightarrow x = (\tilde{x}, 1) = \sum_{\substack{I_\alpha \\ \hat{I}_\alpha \text{ position}}} \lambda_\alpha x^{I_\alpha + i} \Rightarrow x \in P(M_1 \cap M_2).$$

Proof — continued —

(iii) Suppose $0 < \lambda_i < 1 \quad \forall i \in E$.

Construct chains : $\mathcal{C} : \emptyset \subseteq C_1 \subset \dots \subset C_k$
 $\mathcal{D} : \emptyset \subset D_1 \subset \dots \subset D_l$ } height,
 $\lambda(C_i) = r_1(C_i)$
 $\lambda(D_j) = r_2(D_j)$



$$|C_{i+1} \setminus C_i| \geq 2 \quad \forall i$$

$$|D_{j+1} \setminus D_j| \geq 2 \quad \forall j \Rightarrow k, l \leq |E|/2.$$

Since : $\{\chi^u : u \in \mathcal{C} \cup \mathcal{D}\}$ contains $|\text{supp}(z)| = \underline{|E|}$
 $k+l$ vectors total $\Rightarrow k+l = \underline{|E|/2}$

$$\Rightarrow C_k = E, D_l = E$$

$\Rightarrow \mathcal{C} \cup \mathcal{D}$ have at most $|E|-1$ sets. *



